

Modeling with a Parent Function: the Open Box - Practice

Algebra II | Unit 1 | Lesson 9 Practice | TEKS A2.2.A, A2.7.I

Objective: I can build a volume function for an open box, find the contextual domain, and reason about how volume scales.

Section 1 — Recall

1. A sheet 14 in by 6 in has squares of side x cut from each corner. Write the height, length, and width in terms of x .

height: _____ length: _____ width: _____

2. Which side, 14 or 6, limits the domain, and why?

Section 2 — Apply

3. Write the volume function $V(x)$ for the 14-by-6 sheet, and give the contextual domain.

$V(x) =$ _____ Domain: _____

4. Compute $V(1)$ for the 14-by-6 sheet. Show your arithmetic.

$V(1) =$ _____

Section 3 — Challenge

5. A sheet is 24 in by 18 in. Using the pattern "domain limited by the shorter side," find the contextual domain without fully expanding $V(x)$.

Domain: _____

6. For the original 20-by-10 sheet, the max volume was about 192.45 at x about 2.11. For a new sheet exactly half the size in each dimension (10-by-5), predict - without computing - whether the new max will be exactly half, more than half, or less than half of 192.45, and justify using how volume scales with all three dimensions.

Section 4 — Explain It

7. A classmate says: "The contextual domain for any open-box problem should always be $(0, \text{half of the longer side})$, since a bigger sheet means a bigger allowed cut." Is the classmate correct? Explain using the original 20-by-10 sheet.
